Huawei Access Node HEB-ACC-75 Failure Resolution SOP (Critical - ALARM_7345)

Node: HEB-ACC-75 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_7345

Description

This SOP outlines the steps to identify and resolve hardware/environmental faults on the Huawei Access Node HEB-ACC-75, specifically addressing ASIC memory parity errors, line card crashes, and power supply redundancy loss. The Access Node, a critical component of the Hebbal Edge site, plays a vital role in ensuring seamless data transmission between the core network and customer-facing equipment. The failure of this node can lead to significant service disruptions, emphasizing the need for swift and effective resolution. The underlying protocol mechanics of the Huawei Access Node involve the use of advanced ASIC (Application-Specific Integrated Circuit) technology, which enables high-speed data processing and transmission. However, these ASICs are prone to memory parity errors due to the complex interactions between various system components. Furthermore, line card crashes and power supply redundancy loss can occur due to hardware malfunctions or environmental factors such as power outages or extreme temperatures. To mitigate these risks, regular maintenance, monitoring, and prompt resolution of faults are essential.

Pre-requisite Checks

- Verify the site location (Hebbal Edge) and node ID (HEB-ACC-75) to ensure the correct equipment is being addressed.
- Confirm the alarm code (ALARM_7345) and severity (Critical) to understand the nature of the failure.
- Check the redundant link status to ensure there are no other affected components in the network.
- Dispatch remote hands to the Hebbal Edge site for physical access and assistance if required.

Potential Root Causes

Cause	Probability
ASIC memory parity errors due to software or hardware inconsistencies	High
Line card crashes due to hardware malfunctions or power supply issues	Medium
Power supply redundancy loss due to environmental factors or hardware failure	Low

Diagnostic Commands & Expected Outputs

```
Command: display current-configuration | include ASIC, Expected Output: Current ASIC configuration, including settings and status, Description: This command isolates the ASIC configuration, allowing the technician to verify if any settings have been altered or if there are any indications of hardware malfunctions.
```

```
Command: display cpu-usage | include ASIC, Expected Output: CPU usage statistics for the ASIC, including any errors or anomalies, Description: This command provides CPU usage statistics for the ASIC, enabling the technician to identify any potential performance issues or errors that may be contributing to the failure.

Command: display power-supply | include redundancy, Expected Output: Power supply redundancy status, including any issues or warnings, Description: This command isolates the power supply redundancy status, allowing the technician to verify if there are any issues or warnings that may be contributing to the failure.
```

Resolution Steps

Step 1: Step: Reset the line card, Command: reset line-card 1/0/1-2, Description: This command resets the line card, which may resolve any issues related to line card crashes or hardware malfunctions.

Step 2: Step: Restart the power supply, Command: restart power-supply 1/0/1-2, Description: This command restarts the power supply, which may resolve any issues related to power supply redundancy loss or environmental factors.

Step 3: Step: Reconfigure the ASIC, Command: system-view config asic 1 reboot, Description: This command reconfigures the ASIC, which may resolve any issues related to ASIC memory parity errors or software inconsistencies.

Post-Check Verification

```
Command: display system-logs | include ASIC, Expected Output: System logs related to ASIC, including any errors or warnings, Description: This command provides system logs related to the ASIC, enabling the technician to verify if the resolution steps have resolved the issue.

Command: display cpu-usage | include ASIC, Expected Output: CPU usage statistics for the ASIC, including any errors or anomalies, Description: This command provides CPU usage statistics for the ASIC, enabling the technician to verify if the resolution steps have resolved any performance issues or errors.
```

Roll-back Procedures

- Step: Restore the original ASIC configuration, Command: system-view config asic 1 undo reboot save, Description: This command restores the original ASIC configuration, ensuring that any changes made during the resolution process are rolled back.

- Step: Restart the power supply, Command: restart power-supply 1/0/1-2, Description: This command restarts the power supply, ensuring that any changes made during the resolution process are rolled back.